

Basic Principles of Medicine 2

Module: Foundations to Medicine

Please review before the lecture!

Excitable Cells

Drs Stephan Bandelow and Duncan Kirkby

DLA Objectives

- SOM.MK.I.BPM2.1.NB.1.NSCI.0036 describe the influence of the influx of sodium ions on the rising phase of the action potential
- SOM.MK.I.BPM2.1.NB.1.NSCI.0037 describe the influence of the efflux of potassium ions on the repolarizing phase of the action potential
- SOM.MK.I.BPM2.1.NB.1.NSCI.0038 recognize the different time courses of changes in sodium and potassium conductance
- SOM.MK.I.BPM2.1.NB.1.NSCI.0039 explain absolute and relative refractory periods

“Excitable Cells” DLA overview

- **Na⁺ and K⁺ Movements During the Impulse**

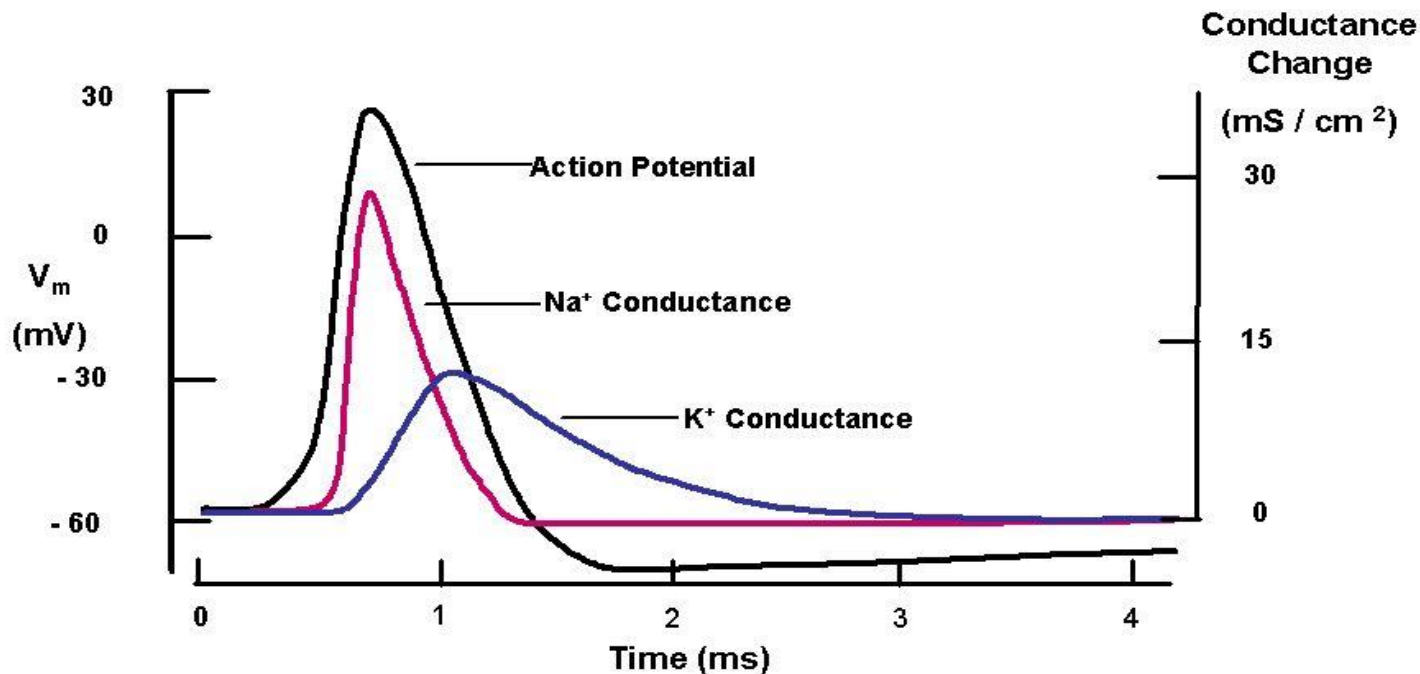
- During subthreshold depolarization at the impulse initiation zone, a few **voltage-gated Na⁺ channels (Na_v)** may open, but depolarization is insufficient to trigger widespread Na_v opening. Efflux of K⁺ via leak channels also electrically offsets the modest entry of Na⁺. Once the depolarization threshold is reached many of the Na_v channels open, and the resulting net influx of positive charge further depolarizes the membrane to open additional Na_v channels. A positive feedback cycle thus rapidly ensues, with Na⁺ cascading inwards along its electrochemical gradient ($V_m - E_{Na}$). The **rising phase** of the action potential thus reflects Na⁺ influx and the membrane depolarizing towards E_{Na} .
- Depolarization also induces the delayed opening of voltage-gated K⁺ channels. The efflux of K⁺ ions along their electrochemical gradient eventually repolarizes the membrane, thus yielding the **falling phase** of the impulse. En route to E_K , the membrane may hyperpolarize (below resting V_m), creating an undershoot. Therefore, action potentials initially depolarize towards E_{Na} , then repolarize and hyperpolarize towards E_K .

- **Changes in g_{Na} and g_K Underlie Action Potentials**

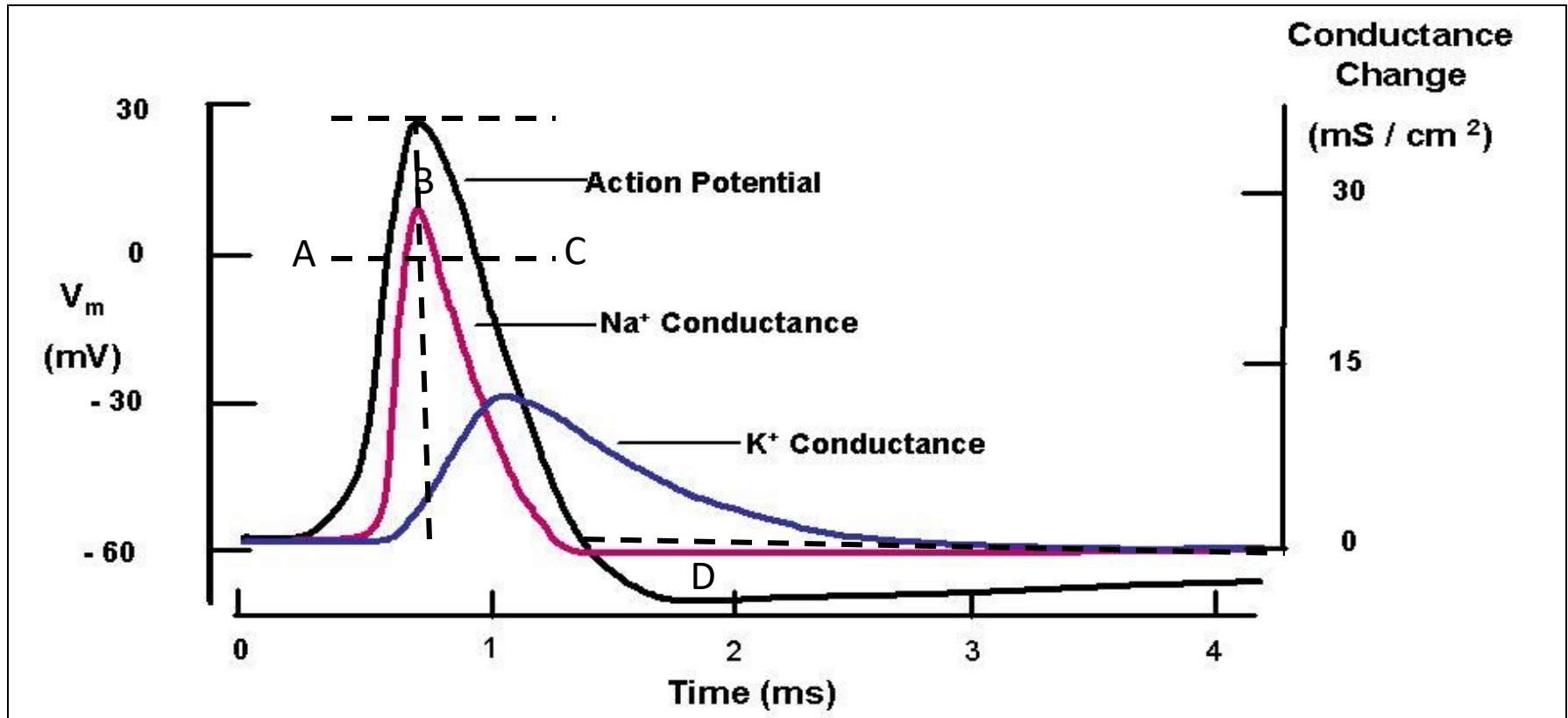
- The Na⁺ conductance (g_{Na}) curve reflects the transient activity of voltage-gated Na⁺ channels during the action potential. Note that g_{Na} falls owing to a depolarization-induced **inactivation** of Na⁺ channels. The g_K curve shows the slower time course of the opening of K⁺ channels during the action potential. The rising phase of the action potential thus reflects rising g_{Na} . The falling phase of the action potential reflects declining g_{Na} coupled with a rising potassium conductance (g_K).

Ionic Basis of Action Potentials

- Action potential reflects temporally offset Na^+ and K^+ conductances
 - Na^+ channels open, then inactivate
 - K^+ channels wait, then open and inactivate more slowly



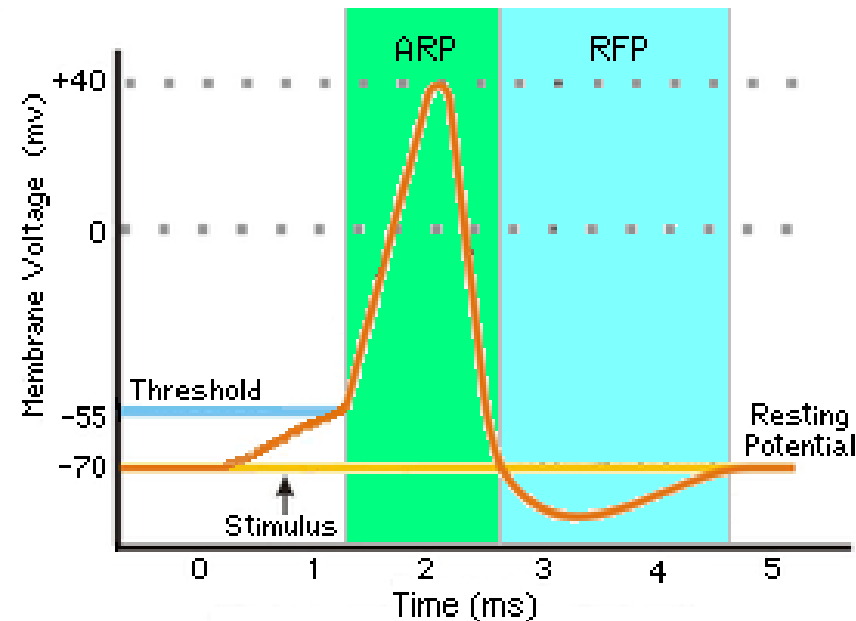
Na⁺ and K⁺: Phases of Action Potentials



- A – Rising phase
- B – Overshoot
- C – Falling phase
- D – After-hyperpolarization

Refractory periods

- Absolute refractory period (ARP)
 - during repolarization phase when inactivation gates are closed
 - no new AP can be initiated or conducted because Na_v channels are blocked (inactivated)
- Relative refractory period (RFP)
 - after absolute refractory period during afterhyperpolarization
 - starts when the Na_v inactivation gates are open again
 - APs initiation is inhibited due to afterhyperpolarization (increased K^+ conductance)
 - but principally APs can again be initiated since Na_v channels have reset (inactivation gates open)



Formative Assessment

A 43-year-old man is brought to the emergency department because of a 30-minute history of muscle spasms. It is suspected that he accidentally ingested a potassium channel blocking agent. Which of the following action-potential-related events is most likely affected by the drug in this patient?

- A. Rising phase
- B. Falling phase
- C. Rising and falling phases
- D. After-hyperpolarization
- E. Falling phase and after-hyperpolarization
- F. Rising phase and after-hyperpolarization

A 43-year-old man is brought to the emergency department because of a 30-minute history of muscle spasms. It is suspected that he accidentally ingested a potassium channel blocking agent. Which of the following action-potential-related events is most likely affected by the drug?

- A. Rising phase
- B. Falling phase
- C. Rising and falling phases
- D. After-hyperpolarization
-  E. Falling phase and after-hyperpolarization
- F. Rising phase and after-hyperpolarization

The falling phase and the after-hyperpolarization reflect potassium currents